

Auxiliary Tasks to Improve Final Prediction

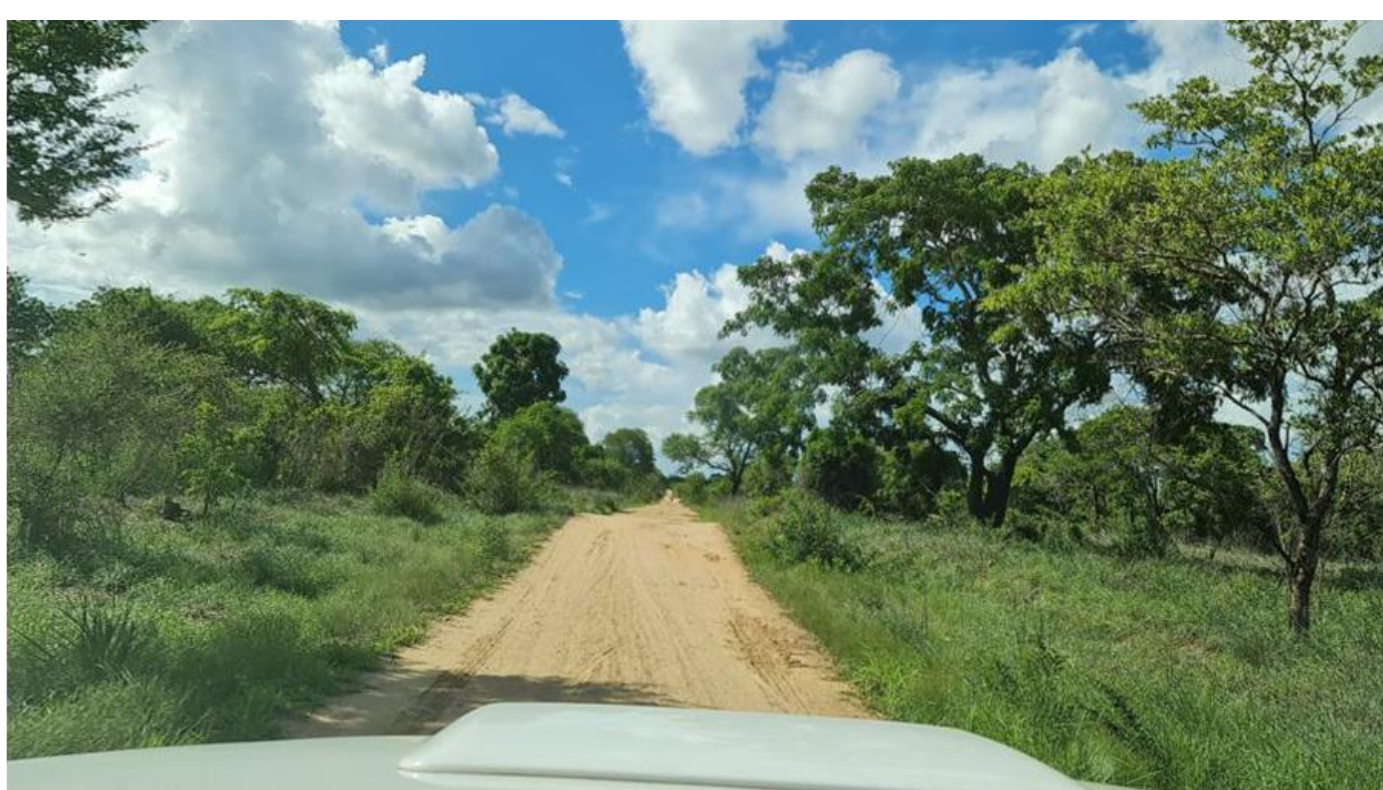
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Overview:

- Many deep learning tasks produce deceptively simple outputs while requiring models to reason through complex intermediate representations. For example:
 - Protein-ligand binding is a binary classification task, but it depends on first predicting protein structure, binding-pocket geometry, and chemical interactions.
 - Image geolocation only outputs coordinates, but it requires understanding of intermediate features such as climate, terrain, geography, and architecture.
- This project asks: how does stacking multiple auxiliary predictions before a final prediction impact model performance?
- Specifically, we explore how image geolocation can be improved by first making auxiliary predictions.

Data:

- All data is sourced from the Open Street View 5M dataset.
- We use 170k train images and 15k eval/test images each.
- Each image is labeled with latitude, longitude, country, climate, land cover, and distance to sea.



Latitude: -15.40137... Longitude: 40.23688...
Country: MZ Climate: 3 Land Cover: 4

Figure 1: Example training image and its labels.



Figure 2: Countries represented in the train set.

Experiments:

- We built 5 architectures, each of which integrate auxiliary predictions differently.
- The architectures are: control, multi-loss, parallel heads, sequential heads, and mixture of experts (MOE).
- The models are constructed from a set of shared components.

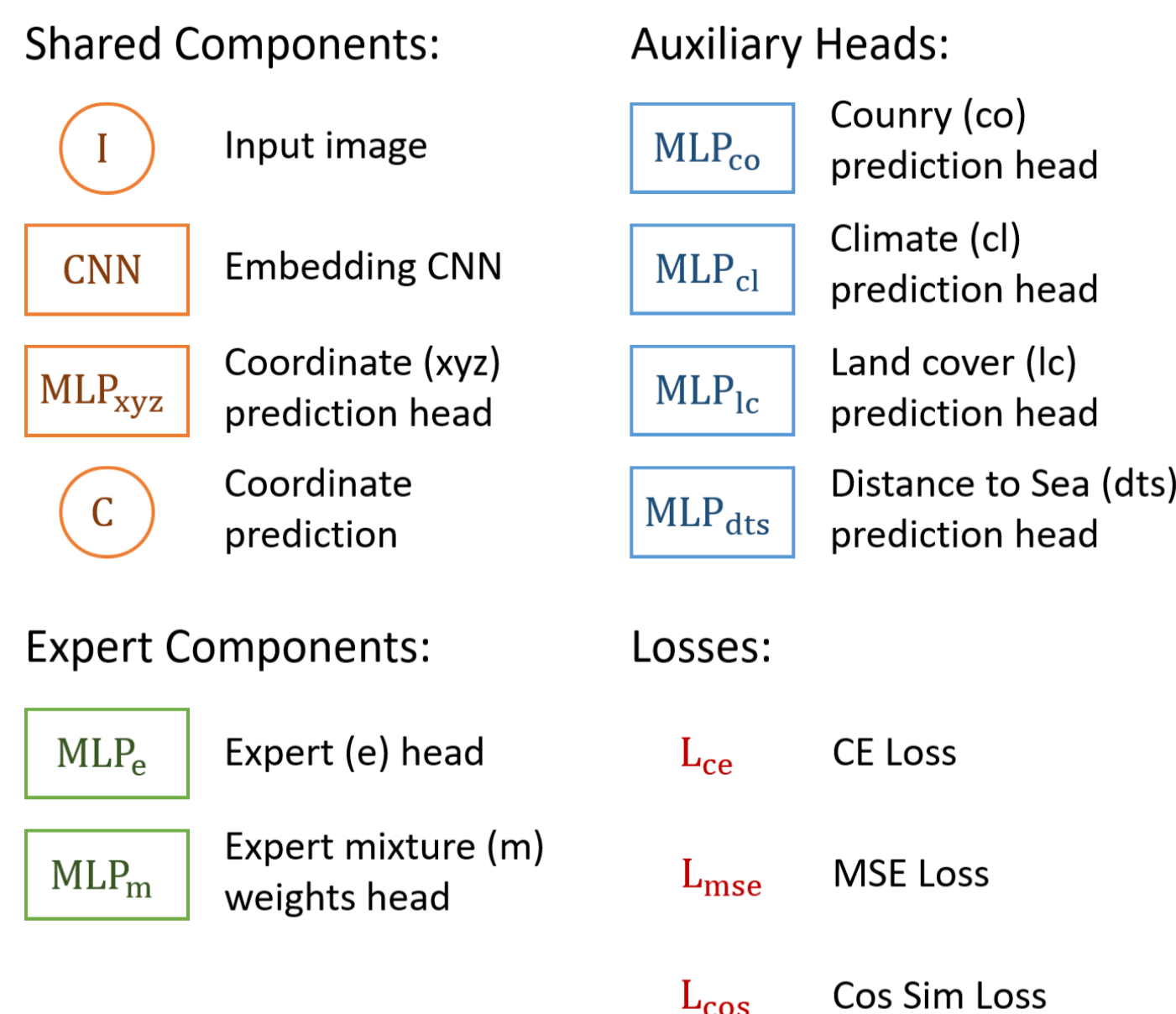


Figure 3: The components used to construct each architecture.

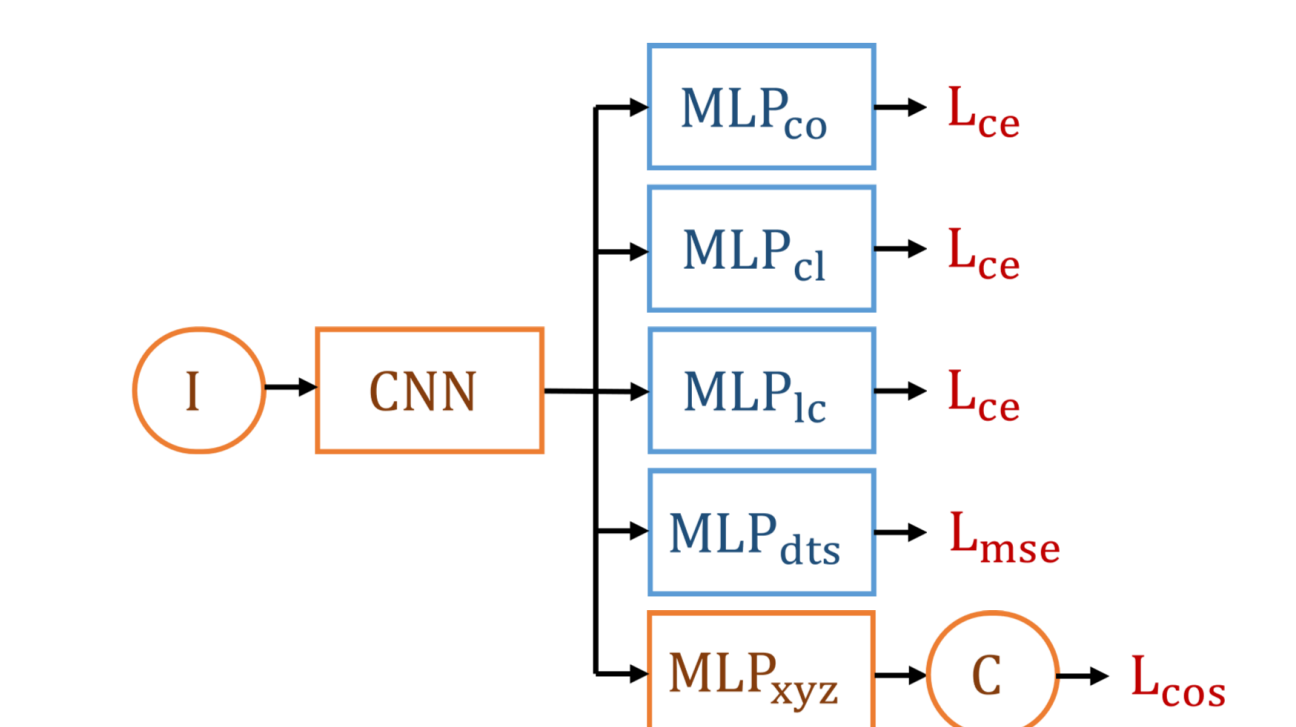
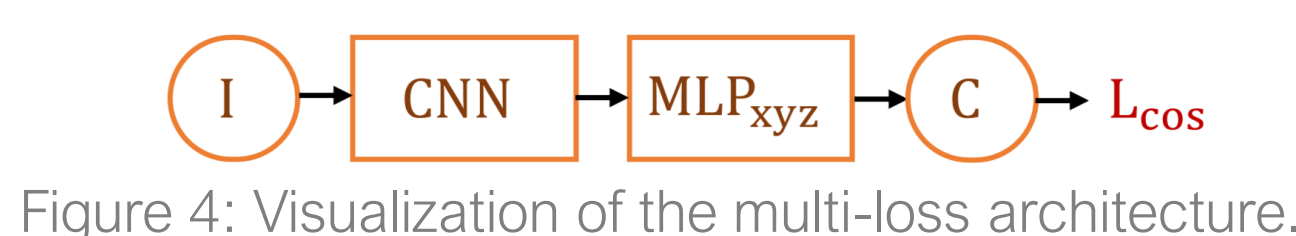


Figure 5: Visualization of the multi-loss architecture.

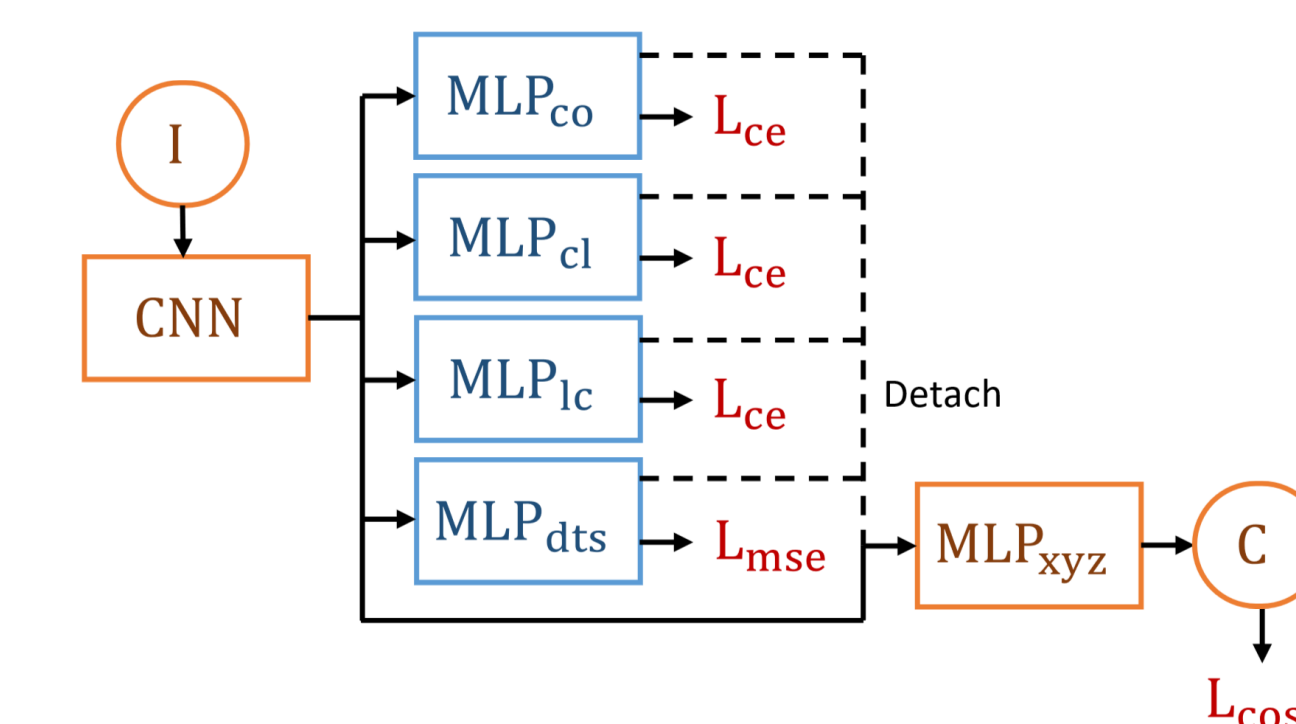


Figure 6: Visualization of the parallel architecture.

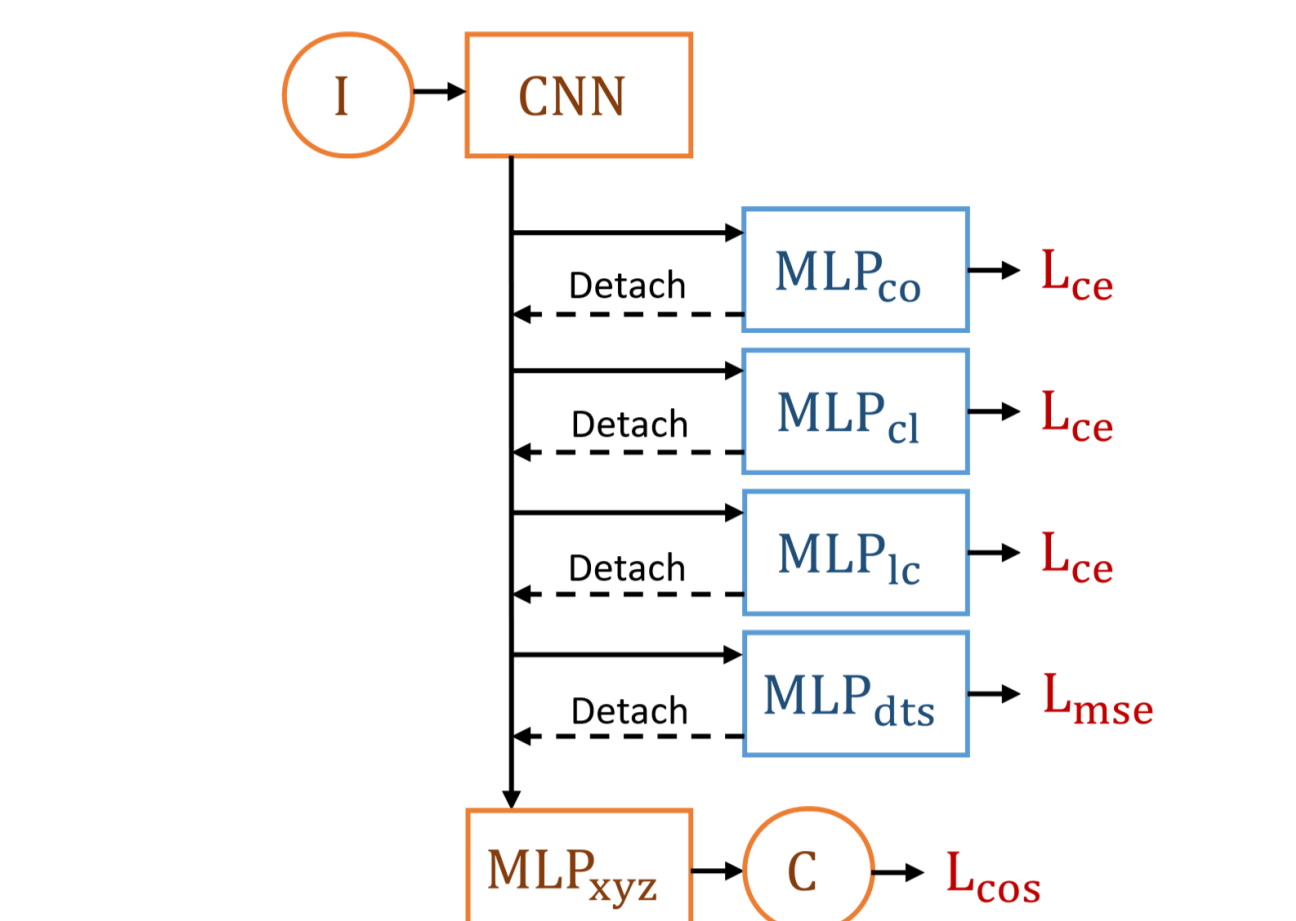


Figure 7: Visualization of the sequential architecture.

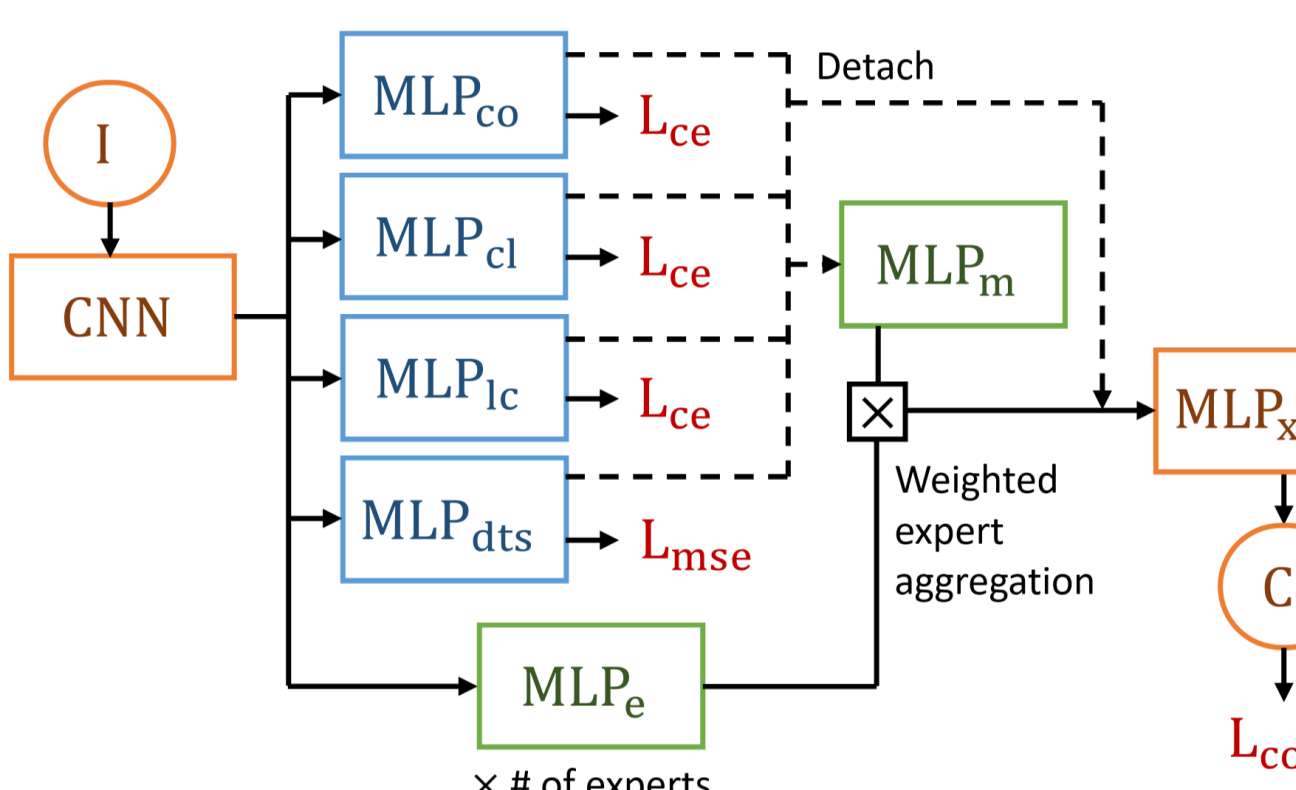
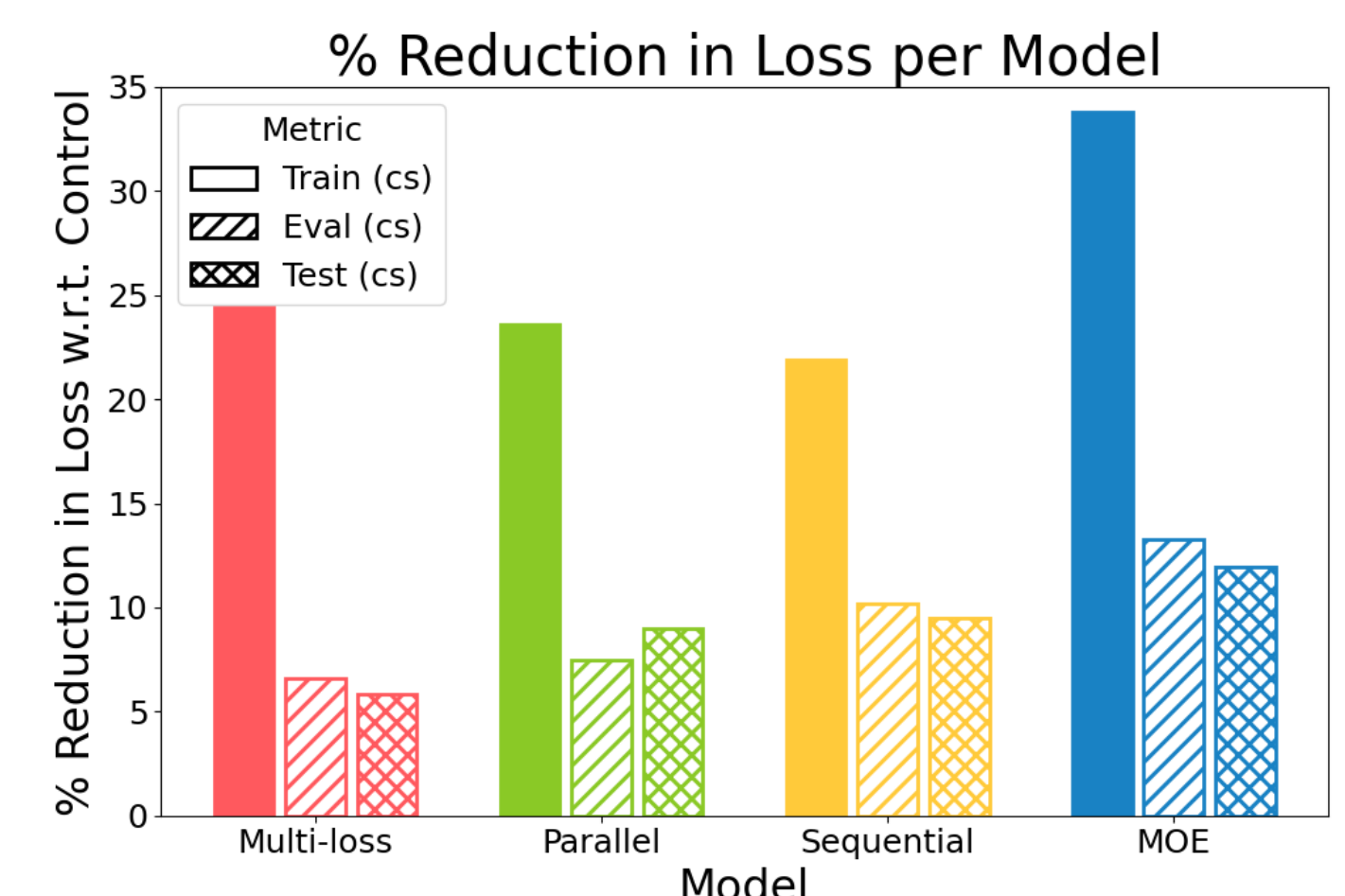
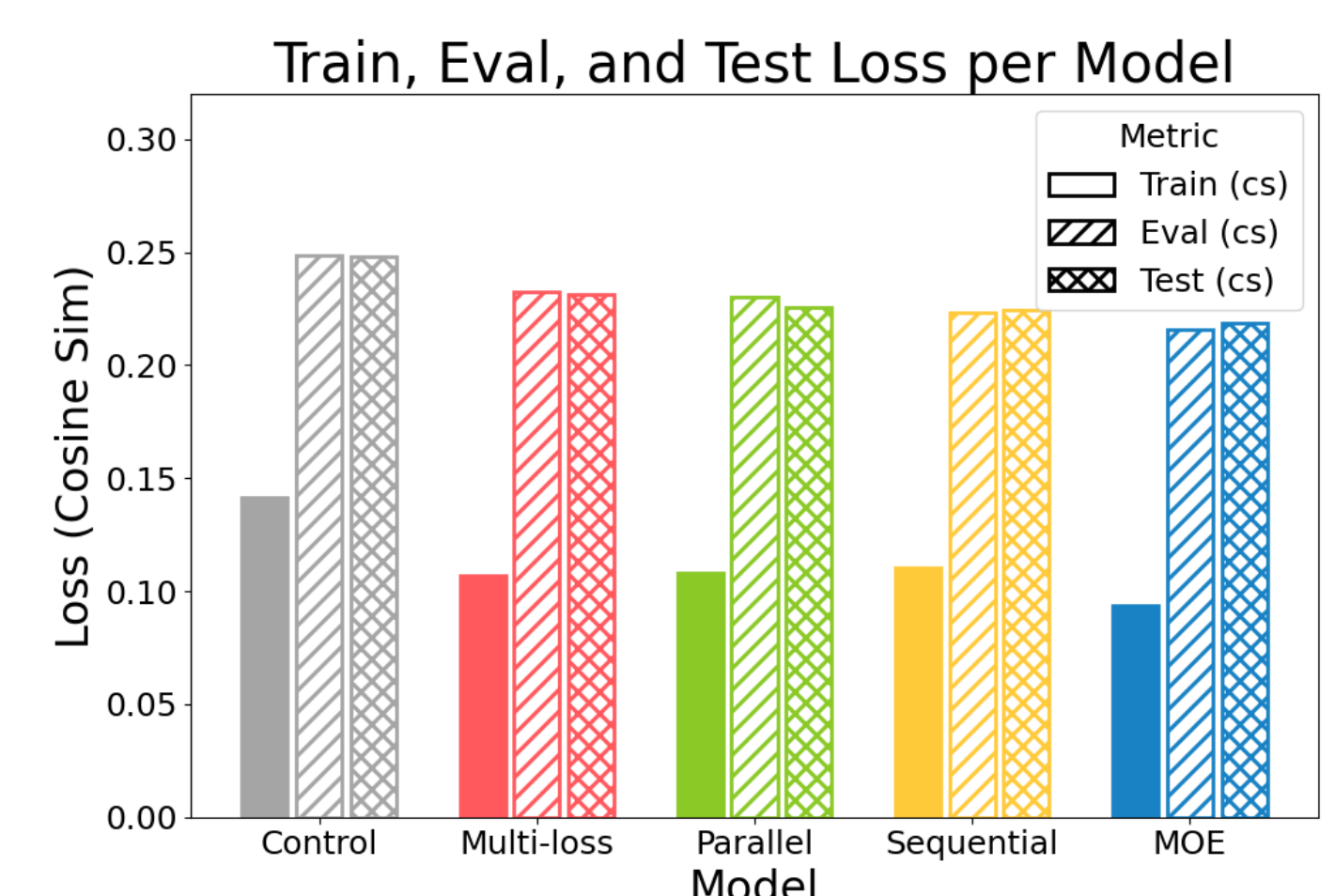
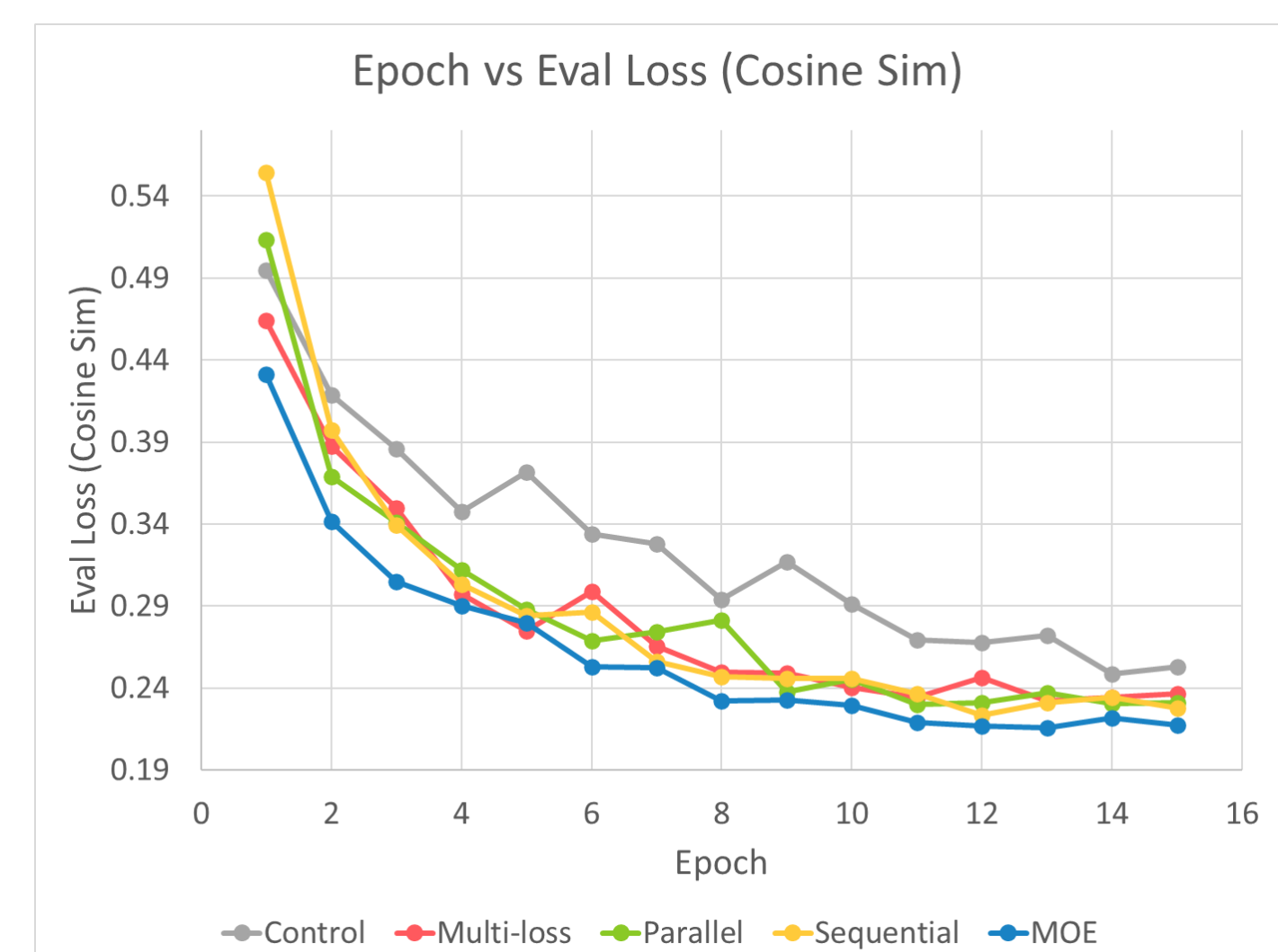
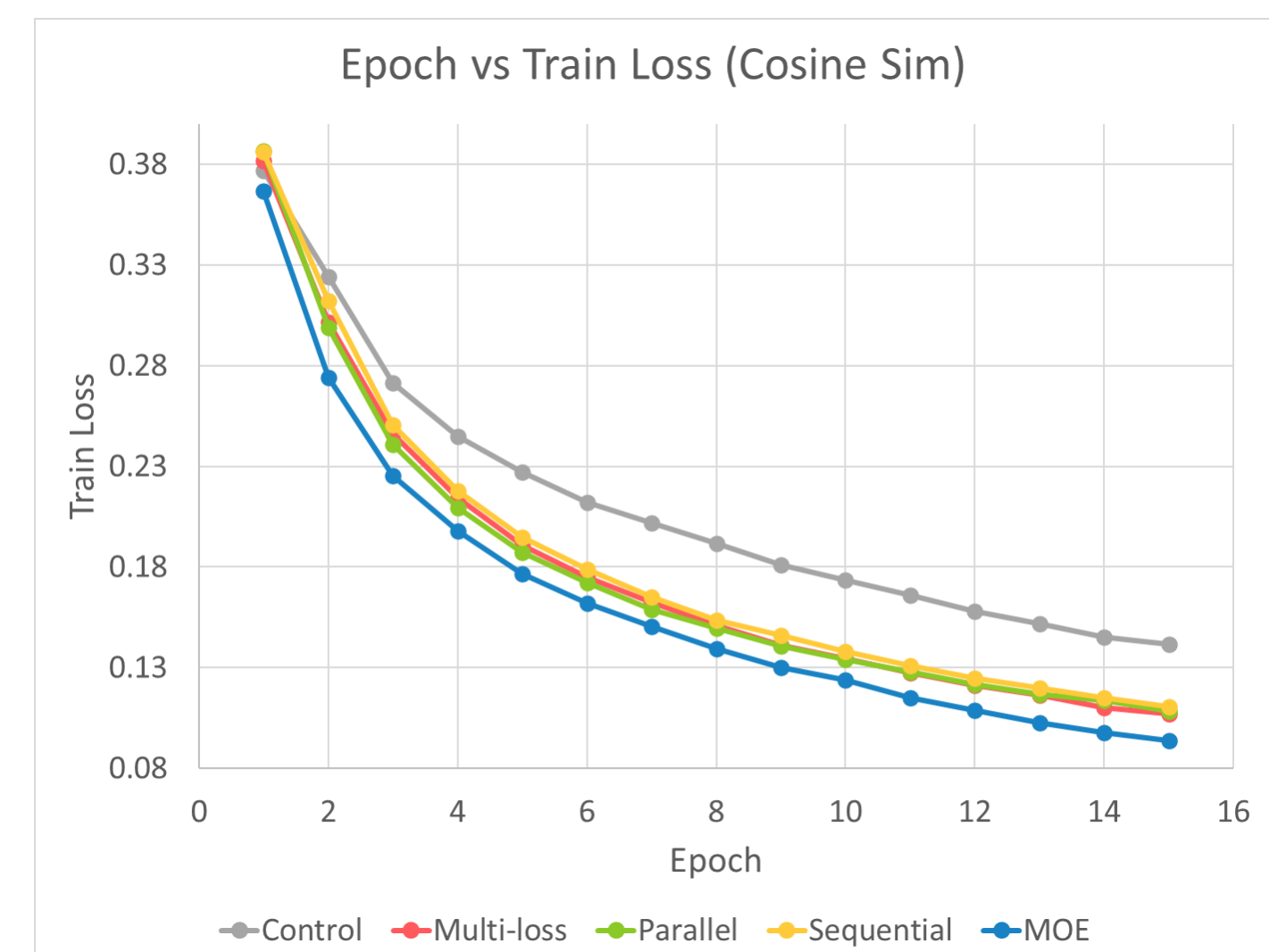


Figure 8: Visualization of the MOE architecture.

Results:

- The control model performed worst, converged slowest, and produced the least stable validation curves.
- The MOE model performed best, converged fastest, and produced the most stable validation curves.
- Validation losses were about 2 times higher than training losses, with the smallest gap for the control model and the largest gap for the MOE model. This is likely due to limited train data.



Works Consulted:

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